

Glaucoma and Cataract Diagnosis from AS-OCT Images Using Ensemble Learning

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Abstract

Glaucoma and cataract are among the leading causes of preventable blindness worldwide, and their co-occurrence in elderly populations complicates clinical decision-making. Traditional diagnostic standards, such as gonioscopy, remain examiner-dependent and limited in reproducibility, whereas anterior-segment optical coherence tomography (AS-OCT) provides high-resolution, quantitative imaging suitable for automated analysis. In this study, we propose a deep learning framework for the four-class classification of AS-OCT images into Normal, Cataract, Glaucoma, and Glaucoma&Cataract. To ensure credible evaluation, we enforce strict patient-level data partitioning across training, validation, ensemble training, and testing sets, thereby avoiding data leakage. Our approach leverages a heterogeneous backbone library of convolutional neural networks, systematically benchmarked under multiple ensemble learning strategies, including statistical aggregation, selection-based methods, and meta-learning. Experimental results on a dataset of 20,144 AS-OCT images from 657 patients demonstrate that ensemble integration consistently improves performance over single models. Logistic regression stacking achieved the highest accuracy (89.16%) and AUC (0.974), surpassing the best individual backbone. Interpretability analysis with Grad-CAM further confirmed that the model's attention aligns with clinically relevant anatomical structures. These findings establish a robust baseline for automated, one-stop triage of glaucoma and cataract and highlight the translational potential of ensemble deep learning in ophthalmic diagnostics.

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CCS Concepts

• Applied computing; • Life and medical sciences; • Health informatics;

Keywords

AS-OCT, Patient-level Stratification, Ensemble Learning, Medical Image Classification

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1 INTRODUCTION

Glaucoma and cataract are leading causes of visual impairment and avoidable blindness worldwide, and their combined burden is rising with population aging [1-2]. These diseases frequently co-occur among elderly adults. Managing either condition in isolation can be unsafe: focusing solely on “angle closure” may overlook lens-driven mechanisms that favor lens surgery, whereas focusing solely on “cataract” risks missing glaucoma patients who require urgent pressure-lowering intervention. These considerations motivate a practical, one-stop, objective triage solution that partitions examinees into four clinically meaningful categories (Normal, Cataract, Glaucoma, Glaucoma & Cataract) to guide downstream care.

Along the diagnostic pathway, gonioscopy remains the reference standard for assessing the anterior chamber angle, yet it is examiner-dependent, subjective, and difficult to quantify [3]. Anterior-segment optical coherence tomography (AS-OCT), by contrast, is non-contact, high-resolution, and parameterizable. It objectively captures cornea, chamber, iris, lens relationships and has demonstrated strong accuracy for detecting angle closure across

studies and meta-analyses [3, 4]. AS-OCT is therefore a natural substrate for building scalable and standardized systems for complex diagnostic scenarios.

In recent years, artificial intelligence (AI) has advanced rapidly, driving significant progress in medical image analysis and bringing substantial convenience to clinical workflows [5–7]. Automated analysis of AS-OCT images has also advanced rapidly: deep learning systems now achieve strong performance for angle-closure detection and show encouraging cross-cohort transferability, while automated cataract-related assessment has seen meaningful progress as well [8–12]. Yet two methodological gaps continue to limit clinical reliability and translation. First, some studies have not enforced patient-level dataset partitioning, allowing multiple images from the same individual to appear in both training and test sets. This data leakage materially inflates reported accuracy and undermines external generalization, which is an evaluation pitfall well documented in medical imaging AI [13]. Second, a single backbone network carries fixed inductive biases that are ill-suited to the multi-structure, multi-scale heterogeneity of AS-OCT images, particularly in real-world Glaucoma & Cataract comorbidity. Few prior automated studies have attempted joint triage of glaucoma and cataract in a single framework.

To address these needs, we develop an AS-OCT framework explicitly tailored to the Normal, Cataract, Glaucoma, Glaucoma & Cataract task under strict patient-level splits and evaluation. Instead of pursuing a single “best” model, we construct a heterogeneous backbone library and systematically compare ensemble strategies to reduce variance, mitigate class imbalance effects, and improve cross-device/center robustness [14, 15]. In doing so, we aim to provide an evaluation-credible and performance-robust baseline that aligns with clinical workflows and advances automated, one-stop triage for glaucoma and cataract.

Our contributions are summarized as following three points:

This study is the first to systematically utilize AS-OCT images for the simultaneous investigation of two diseases, cataract and glaucoma, making it more aligned with real-world clinical diagnostic scenarios.

A strict patient-level data partitioning strategy was adopted to eliminate the risk of data leakage, thereby ensuring the objectivity of model performance evaluation and the credibility of results.

This study systematically compared the performance of various mainstream ensemble learning strategies, providing a reliable reference baseline for subsequent method optimization and clinical translation of this task.

2 RELATED WORK

2.1 Deep Learning Classification Models for AS-OCT Images

In recent years, intelligent diagnosis research for ophthalmic diseases based on AS-OCT images has made significant progress. Various deep learning methods have emerged for the classification tasks of glaucoma and cataract, two common blinding eye diseases.

In the direction of glaucoma classification, research predominantly focuses on the angle-closure mechanism: Hao et al. (2022) developed a multi-view volume and temporal difference network

combining CNN-LSTM architecture for screening angle-closure glaucoma from AS-OCT videos, achieving high sensitivity and specificity on specific datasets [16]; Rasel et al. (2024) confirmed the effectiveness of OCT-based 2D/3D CNN algorithms in glaucoma detection [17]. In the direction of cataract classification, research places greater emphasis on the quantitative analysis of lens opacity: Zhang et al. (2022) proposed a clinical global-local feature framework for nuclear cataract classification in AS-OCT images, which enhances the representation of both coarse anatomical structures and fine lesion regions, reflecting the trend of leveraging multi-scale and clinically inspired feature design in cataract analysis [18].

While prior studies have advanced the field substantially, most were not designed for broader multi-class settings that jointly consider glaucoma and the clinical co-occurrence of glaucoma and cataract. In addition, many reports evaluate a single deep model, which can be more sensitive to AS-OCT’s small inter-class margins and large intra-class variability.

2.2 Ensemble Learning in Medical Image Classification

In recent years, ensemble learning has become a key strategy for enhancing the performance, robustness, and generalization ability of deep learning-based medical image classification models, effectively addressing challenges such as limited annotated data and class imbalance [19].

Müller et al. (2022) evaluated boosting ensemble, stacking ensemble, and cross-validation-based bagging ensemble on four datasets based on nine CNN architectures, finding that the stacking ensemble achieved the greatest improvement in F1-score, and simple pooling functions performed excellently [19]; Qian et al. (2023) proposed a voting-stacking ensemble strategy, using three Inception models as base learners and logistic regression as the meta-learner, achieving 100% accuracy on three cervical cytology datasets [20]; Gautam et al. (2024) designed a weighted ensemble model composed of ResNet-152, DenseNet169, and EfficientNet-B7, combined with a novel weight optimization scheme, achieving 97.23% accuracy on the LIDC-IDRI lung cancer dataset [21]; Asiri et al. (2023) fine-tuned five pre-trained ViT models for brain tumor classification, among which ViT-b32 achieved an accuracy of 98.24%, outperforming CNN-based methods [22]; Zheng et al. (2022) combined transfer learning with ensemble pruning, fine-tuning six CNN models and then selecting the four best models for weighted voting, achieving 98.90% accuracy on the BreakHis breast histopathology dataset [23].

These studies fully demonstrate the value of ensemble learning in medical image analysis, but most focus on areas such as lung cancer, brain tumors, and breast diseases

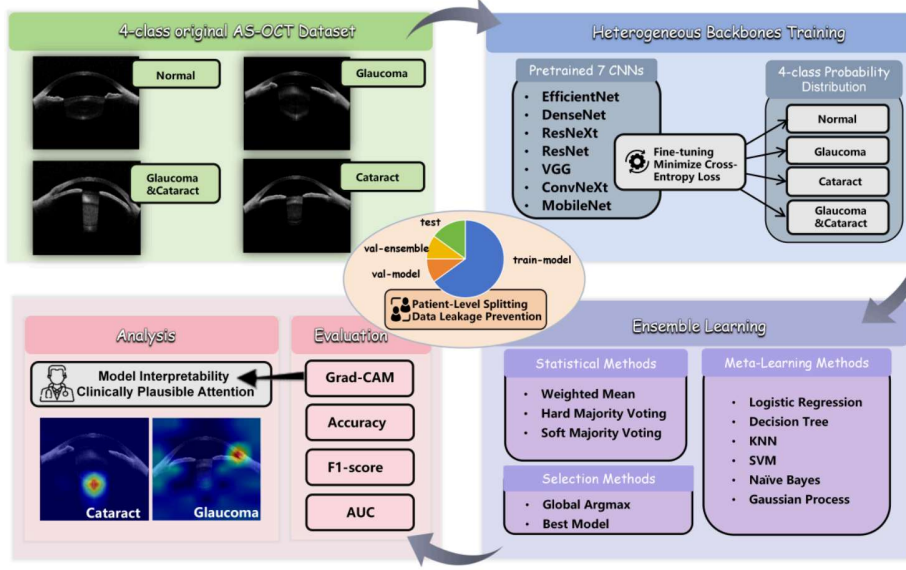


Figure 1: Overall Framework of Glaucoma and Cataract Diagnosis from AS-OCT Images Using Ensemble Learning

3.1 Problem Formulation

We study four-class diagnosis from AS-OCT B-scans. Let the image space be $\mathcal{X} \subset \mathbb{R}^{H \times W}$ and define the 4-class probability simplex

$$\Delta^4 = \left\{ \mathbf{p} \in \mathbb{R}_{\geq 0}^4 : \sum_{c=1}^4 p_c = 1 \right\} \quad (1)$$

The label set is $\mathcal{Y} = \{\text{Normal}, \text{Cataract}, \text{PACG}, \text{PACG\&Cataract}\}$. Given a dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ collected from multiple subjects, the goal is to learn a classifier $f_\theta : \mathcal{X} \rightarrow \Delta^4$ that outputs the class posteriors $\mathbf{p}(\mathbf{x}) = f_\theta(\mathbf{x})$.

To ensure clinical generalization and avoid patient-level information leakage, we adopt a subject-exclusive four-way split $[\mathcal{D}_{train}, \mathcal{D}_{val}, \mathcal{D}_{ens}, \mathcal{D}_{test}]$, where all images from a given subject appear in exactly one subset. Backbones are trained on $\mathcal{D}_{train}$, model selection is performed on $\mathcal{D}_{val}$, all ensemble parameters are fit only on $\mathcal{D}_{ens}$, and final performance is reported exclusively on $\mathcal{D}_{test}$.

To mitigate single-model inductive bias on heterogeneous AS-OCT structures, we train M base models $\{B_k\}_{k=1}^M$, each producing raw softmax posteriors $\mathbf{p}_k(\mathbf{x}) \in \Delta^4$. The ensemble learns a fusion mapping $F : (\Delta^4)^M \rightarrow \Delta^4$ on $\mathcal{D}_{ens}$, and the final prediction for a test image $\mathbf{x}$ is

$$\hat{y} = \arg \max_{c \in \mathcal{Y}} [F(\mathbf{p}_1(\mathbf{x}), \dots, \mathbf{p}_M(\mathbf{x}))]_c \quad (2)$$

Detailed introductions of backbones and ensemble models will be discussed below.

3.2 Backbones Training

Our base layer consists of M complementary 2D classifiers, such as ResNet, DenseNet, EfficientNet, and ConvNeXt. Each model is first initialized with weights pre-trained on ImageNet and then equipped with a new four-class linear head featuring a softmax activation function for our specific task. Subsequently, the entire network is fine-tuned end-to-end on $\mathcal{D}_{train}$. During this fine-tuning process, we handle class imbalance robustly by utilizing a cross-entropy loss function with class weights:

$$\min_{\theta} \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}_{train}} \left[- \sum_{c=1}^4 1\{y=c\} w_c \log p_c(\mathbf{x}) \right], \quad w_c = \frac{1/\text{freq}(c)}{\sum_{c'=1}^4 1/\text{freq}(c')} \quad (3)$$

For each backbone B_k , we select and retain the checkpoint that achieves the best accuracy score on the validation set $\mathcal{D}_{val}$.

3.3 Ensemble Models

Let $\mathbf{p}_1(\mathbf{x}), \dots, \mathbf{p}_M(\mathbf{x})$ denote the raw base posteriors for input $\mathbf{x}$. Fusion parameters are fit only on $\mathcal{D}_{ens}$. We systematically evaluate three families of ensemble methods.

Family A: Statistical Aggregation

Hard Voting. The final prediction is the class that receives the most votes.

$$\hat{y}_{hard}(x) = \arg \max_{c \in \mathcal{Y}} \sum_{k=1}^M \mathbb{I}(\arg \max_{c'} \mathbf{p}_k(x)_{c'} = c) \quad (5)$$

where $\mathbb{I}(\cdot)$ is the indicator function. The output can be converted to a probability vector by dividing the vote count for each class by M .

Weighted Mean. To reflect model quality, assign weights via negative log-likelihood (NLL) on $\mathcal{D}_{ens}$:

$$w_k \propto \exp(-NLL_k(\mathcal{D}_{ens})), \quad \sum_{k=1}^M w_k = 1, \quad (6)$$

$$\mathbf{p}_{wavg}(x) = \sum_{k=1}^M w_k \mathbf{p}_k(x)$$

Family B: Selection and Gating

Instead of combining all models, this family selects a single model's prediction based on certain criteria.

Global Argmax. For each individual sample x , this method selects the prediction from the model B_{k^*} that exhibits the highest posterior probability.

$$k^*(x) = \arg \max_k \left(\max_{c \in \mathcal{Y}} \mathbf{p}_k(x)_c \right), \quad \mathbf{p}_{sel}(x) = \mathbf{p}_{k^*(x)}(x) \quad (7)$$

Best Model Selection. This approach identifies the single best-performing base model on the entire ensemble dataset $\mathcal{D}_{ens}$. This representative model is then used for all predictions on the test set.

$$k^* = \arg \max_k \{ \text{Accuracy}_k(\{\mathcal{D}\}_{ens}), \quad \mathbf{p}_{best}(x) = \mathbf{p}_{\{k^*\}}(x) \} \quad (8)$$

Family C: Stacking

Stacking treats the outputs of the base models as features for training a second-level model, or meta-learner. We first concatenate the posterior vectors from all M base models to form a meta-feature vector $\mathbf{z}(x) = [\mathbf{p}_1(x); \dots; \mathbf{p}_M(x)] \in \mathbb{R}^{4M}$.

A lightweight meta-learner

$$g_\phi: \mathbb{R}^{4M} \rightarrow \Delta^4 \quad (9)$$

is trained on the dataset $\{(z(x_i), y_i) | (x_i, y_i) \in \mathcal{D}_{ens}\}$. We explore a variety of meta-learners, including L2-regularized Logistic Regression, Support Vector Machines (SVM), Decision Trees, K-Nearest Neighbors, Gaussian Naive Bayes, and Gaussian Process classifiers. The final stacked prediction is: $\mathbf{p}_{stk}(x) = g_\phi(\mathbf{z}(x))$.

4 DATASET AND PREPROCESSING

The data is a retrospective collection spanning February 2022 to December 2024. All AS-OCT images were acquired using a Swept-Source OCT device, the Tomey CASIA2. This device operates at a wavelength of 1310 nm, with a scanning speed of 50,000 scans per second, providing an axial resolution of $<10 \mu\text{m}$ in tissue. The acquisitions were performed by trained ophthalmic technicians following a standardized protocol. Each eye was scanned using the "Radial" scan mode, which captures 16 cross-sectional B-scans centered on the corneal apex. During the examination, patients were instructed to fixate on an internal target to minimize eye movement.

Subjects were meticulously categorized into four distinct groups based on comprehensive clinical examinations: Normal, Cataract,

Glaucoma, and Glaucoma&Cataract. Exclusion criteria for the single-pathology and normal groups included any history of other significant ocular conditions, previous intraocular surgery, or ocular trauma. Ultimately, the final dataset comprises 1259 folders with 20144 images from 657 subjects as shown in Table 1.

All AS-OCT images were uniformly resampled to 224x224 pixels before being input into the model, and normalized using the pre-trained mean and standard deviation from ImageNet. We enforced a patient-level split: all AS-OCT images belonging to the same subject (including both left and right eyes) were treated as a single unit and exclusively assigned to one split set.

Table 1: Dataset distribution

Categories	Normal	Cataract	Glaucoma	Glaucoma&Cataract	Total
Subjects	195	284	89	89	657
Eyes	379	537	171	172	1259
Images	6064	8592	2736	2752	20144

Table 2: Performance of Backbones on the Test Set

Models	Accuracy	F1-Score	AUC
ResNet50 [24]	0.8856	0.8796	0.9681
ResNeXt50 [25]	0.8753	0.8693	0.9639
DenseNet169 [26]	0.8650	0.8570	0.9557
EfficientNet_B4 [27]	<u>0.8781</u>	<u>0.8754</u>	<u>0.9680</u>
Vgg16 [28]	0.8428	0.8358	0.9669
ConvNeXt_tiny [29]	0.8703	0.8634	0.9647
MobileNet_v2 [30]	0.8663	0.8614	0.9604

* Bold numbers indicate best performance and the underlined numbers are the second best results.

Table 3: Performance of Ensemble Strategies on the Test Set

Ensemble Method	Category	Accuracy	F1-Score	AUC
Soft Voting	Statistical Method	0.8840	0.8770	0.9705
Hard Voting	Statistical Method	0.8815	0.8741	0.9527
Weighted Mean	Statistical Method	0.8841	0.8772	0.9706
Global Argmax	Selection Method	0.8831	0.8755	0.9695
Best Model	Selection Method	<u>0.8881</u>	<u>0.8815</u>	0.9568
Decision Tree	Meta Learning	0.8644	0.8562	0.9084
KNN	Meta Learning	0.8781	0.8695	0.9432
SVM	Meta Learning	0.8872	0.8787	0.9642
Naive Bayes	Meta Learning	0.8688	0.8700	0.9559
Gaussian Process	Meta Learning	0.8869	0.8790	<u>0.9733</u>
Logistic Regression	Meta Learning	0.8916	0.8842	0.9740

* Bold numbers indicate best performance and the underlined numbers are the second best results.

5.3 Performance of Ensemble Models

To further improve robustness and predictive accuracy, we systematically evaluated 11 ensemble learning strategies, including statistical aggregation, selection-based methods, and meta-learning approaches.

As summarized in Table 3, ensemble methods significantly boosted performance compared with individual backbones. In particular, logistic regression as a meta-learner achieved the highest accuracy (89.16%) and AUC (0.9740), outperforming the strongest single model (ResNet50) by $\sim 0.6\%$ in accuracy and 0.59% in AUC. This demonstrates the effectiveness of logistic regression in learning complementary information from diverse base models.

Interestingly, simple statistical methods such as mean averaging and soft voting also achieved accuracy and AUC comparable to, or slightly exceeding, the best single backbone. This underscores the general effectiveness of lightweight aggregation strategies in medical ensemble learning. Besides, Gaussian Process meta-learning

also produced an AUC of 0.9733, ranking second only to logistic regression, reflecting its strong ability to leverage

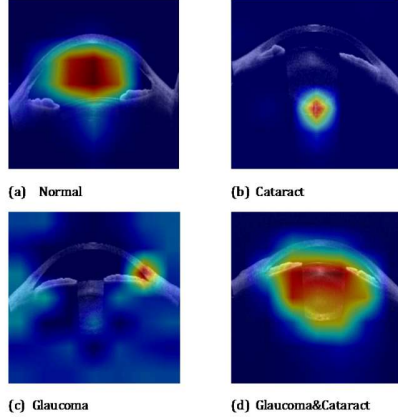


Figure 2: Grad-CAM at output layers for Normal, Cataract, Glaucoma, Glaucoma&Cataract classes

5.5 Ablation Study

To further assess the contribution of model diversity and ensemble scale, we conducted an ablation experiment by progressively varying the number of base models participating in the stacking ensemble. Specifically, we compared three configurations of the logistic regression-based meta-learner: (1) the Top-3 models, selected according to validation accuracy; (2) the Top-5 models; and (3) Ours, which integrates all seven base models. The results are summarized in Table 4.

As shown in Table 4, model performance steadily improved as more base learners were incorporated, demonstrating that multi-model complementarity effectively enhances overall robustness and discriminative power. Compared with the Top-3 and Top-5 ensembles, the full seven-model configuration achieved the highest performance in all assessment criteria. This indicates that each backbone contributes unique feature representations that the meta-learner can exploit to form a more comprehensive decision boundary. Notably, although adding additional models increases computational cost, the gain in stability and generalization justifies this trade-off for clinical deployment scenarios requiring reliable diagnostic confidence.

6 CONCLUSION

We developed and comprehensively evaluated a deep learning framework for automated classification of AS-OCT images into four categories: Normal, Cataract, Glaucoma, and Glaucoma&Cataract.

The framework integrates a heterogeneous set of advanced convolutional backbones and systematically benchmarks eleven ensemble learning strategies to maximize diagnostic performance and robustness.

Our results demonstrate that ensemble integration effectively enhances classification robustness and interpretability, suggesting strong translational potential for clinical use. Beyond model performance, a crucial next step is integration into clinical workflows and telemedicine settings. The proposed system can be embedded within ophthalmic imaging pipelines to provide automated pre-screening and diagnostic triage, offering immediate feedback to clinicians and reducing their manual workload. In community or primary care contexts where ophthalmologists are scarce, the framework could also support teleophthalmology by enabling remote AS-OCT interpretation, thereby facilitating early detection and referral of cataract and angle-closure glaucoma cases.

Furthermore, although the dataset covers a broad range of disease severities, its scope is currently limited to a single imaging device and population. Future work will expand dataset diversity across different anterior-segment OCT platforms and multi-center cohorts, which is essential to enhance the model’s cross-device generalization and population-level fairness. Incorporating data from diverse ethnicities, age distributions, and acquisition protocols will help the system adapt to real-world clinical variability and promote its deployment in large-scale screening programs.

In summary, our ensemble-based framework provides a solid technical foundation for automated ophthalmic diagnosis, and future extensions toward multi-device generalization and clinical workflow integration will further advance its practical impact in both hospital and telemedicine scenarios.

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